

PolyAgora CTO Specification

ADD Evolution: From Classical ADD → Fixed Star ADD-lite

1. Purpose

This document formalizes the transition from:

- the original Asset-Driven Decay (ADD) framework, to:
- the new Fixed Star-informed ADD-lite architecture.

The purpose of ADD-lite is NOT:

- return prediction,
- alpha generation,
- regime classification.

Its purpose is:

measuring degeneration of admissible future maneuverability.

This reframes ADD from:

collapse-risk estimation

toward:

topology deformation and recoverability degradation detection.

This aligns ADD with:

- Fixed Star logic,
 - block ordering,
 - MRTP,
 - admissibility,
 - maneuverability under changing regime geometry.
-

2. Classical ADD (Version 1)

Original Thesis

“Asset growth without value creation breeds fragility.”

Classical ADD measured divergence between:

- asset-price optimization,
and:
- intrinsic value creation.

The original ADD architecture already established:

- semantic polygon,
- analytic polygon,
- synthetic polygon convergence.

This remains foundational.

3. Classical ADD Formula

Original structure:

$$ADD_i = A \times \left(-\frac{\Delta R}{R}\right) \times \frac{1}{D} \times F$$

Where:

- A = Asset dependence
- $-\Delta R/R$ = Return decay
- $1/D$ = Inverse diversification
- F = Fragility

The output:

- quantified structural decay,
- measured collapse pressure,
- identified divergence between price and substance.

This was already philosophically aligned with PolygonEye collapse logic.

4. Limitation of Classical ADD

Classical ADD still remained conceptually close to:

- fragility scoring,
- systemic stress estimation,
- macro-collapse indicators.

It did NOT yet fully integrate:

- Fixed Star ordering,
- admissibility geometry,

- recoverability,
- trajectory degeneration,
- MRTP steering logic.

This becomes the purpose of ADD-lite.

5. Fixed Star Insight

The Fixed Star architecture revealed a deeper structural layer:

The key problem is not merely:

- “which state comes next,”

but:

whether future maneuverability itself is degrading.

This introduces:

- admissibility collapse,
- narrowing corridors,
- synchronization pressure,
- weakening recoverability,
- ordering instability.

The core insight:

superficially similar present states may possess radically different future maneuverability depending on the trajectories that produced them.

This is fundamentally non-Markov.

6. ADD-lite — New Definition

Definition

ADD-lite is:

a low-frequency topology deformation field measuring compression of admissible future maneuverability under increasing synchronization, corridor narrowing, and recoverability degradation relative to the active Fixed Star block structure.

7. Philosophical Shift

Classical ADD

Measured:

- fragility,
- asset decay,
- divergence from value.

ADD-lite

Measures:

degeneration of viable future trajectories.

This is the critical shift.

8. Markov vs ADD-lite

Markov/HMM

Measures:

- state-transition probability.

Core assumption:

$$P(X_{n+1} \mid X_n, \dots, X_0) = P(X_{n+1} \mid X_n)$$

Meaning:

- the current state summarizes relevant history.
-

ADD-lite

Assumes:

- trajectory geometry matters,
- recoverability matters,
- ordering may destabilize,
- transition laws themselves may deform.

ADD-lite therefore measures:

shrinking admissible future-space.

9. ADD-lite Structural Segments

ADD-lite should initially contain 6 low-frequency segments.

C — Corridor Compression

Measures narrowing admissible continuation paths.

Possible proxies:

- liquidity concentration,
- crowding,
- participation narrowing,
- factor concentration.

Interpretation:

- fewer viable continuation paths remain.
-

S — Synchronization Stress

Measures dangerous alignment of trajectories.

Possible proxies:

- cross-asset correlation spikes,
- one-factor dominance,
- positioning synchronization.

Interpretation:

- local optimization becoming globally fragile.
-

B — Breadth Degeneration

Measures deterioration of internal participation.

Possible proxies:

- advance/decline deterioration,
- concentration,
- shrinking diffusion.

Interpretation:

- apparent trend supported by narrowing structure.
-

R — Recoverability Degradation

Measures loss of maneuverability after perturbation.

Possible proxies:

- rebound weakness,
- drawdown persistence,
- volatility clustering.

Interpretation:

- future recovery corridors shrinking.
-

P — Path Dependence Concentration

Measures dependence on continuation of narrow narratives.

Possible proxies:

- thematic concentration,
- leverage concentration,
- duration dependence,
- carry dependence.

Interpretation:

- system increasingly dependent on one continuation trajectory.
-

F — Fragility Asymmetry

Measures nonlinear shock sensitivity.

Possible proxies:

- skew stress,
- convexity stress,
- liquidity gap risk.

Interpretation:

- small shocks capable of producing disproportionate deformation.
-

10. ADD-lite Formula

Conceptual structure:

$$ADD_{lite} = w_C C + w_S S + w_B B + w_R R + w_P P + w_F F$$

Weights:

- frozen initially,
 - low-dimensional,
 - manually interpretable,
 - no optimization against future returns.
-

11. Relationship to Fixed Star Structure

ADD-lite does NOT directly estimate:

- returns,
- regimes,
- prices.

Instead:

it measures:

weakening coherence of the admissible ordering toward the Fixed Star.

Operational interpretation:

- block coherence weakening,
 - corridor narrowing,
 - ridge tension increasing,
 - recoverability deteriorating.
-

12. Runtime Role

ADD-lite is NOT:

- an alpha engine,
- a regime predictor,
- a trading signal.

ADD-lite functions ONLY as:

admissibility and recoverability deformer.

13. Admissibility Deformation

Primary operational form:

$$A_i^* = A_i \cdot (1 - ADD_{lite})$$

Meaning:

- rising ADD-lite compresses admissibility.
-

14. Recoverability Deformation

Secondary form:

$$R_i^* = R_i \cdot (1 - ADD_{lite})$$

Meaning:

- recoverability expectations shrink under topology degeneration.
-

15. Runtime Effects

When ADD-lite rises:

- steering becomes more conservative,
 - concentration reduced,
 - veto sensitivity increases,
 - buffers treated more cautiously,
 - MRTP escalates earlier,
 - maneuverability prioritized over local optimization.
-

16. Historical Validation Methodology

Historical replay is used ONLY for:

structural sanity validation.

Questions:

- Did ADD-lite rise before rupture windows?
- Did synchronization rise before instability?
- Did recoverability shrink before crises?
- Did corridor compression precede breakdown?

NOT:

- maximizing Sharpe,
 - fitting returns,
 - optimizing parameters.
-

17. Forward Validation

True validation must occur through:

forward shadow deployment.

Run:

- baseline $v76\alpha$,
- $v76\alpha$ + ADD-lite,

side-by-side.

Track:

- admissibility changes,
 - recoverability behavior,
 - steering stability,
 - crisis handling,
 - false alarms,
 - maneuverability preservation.
-

18. Guardrails

Forbidden:

- recursive meta-learning,
- online fitting,
- Sharpe optimization,
- direct trading by ADD-lite,
- high-frequency updates,

- unconstrained Kelly amplification.

ADD-lite must remain:

- low-frequency,
 - interpretable,
 - topology-oriented,
 - deformation-focused.
-

19. Strategic Positioning

The objective is NOT:

better regime prediction.

The objective is:

better handling of unstable transition geometry.

This differentiates PolyAgora from:

- Markov allocators,
 - HMM systems,
 - probabilistic regime-switching frameworks.
-

20. Final CTO Direction

Freeze:

- V7.6 α steering kernel.

Build:

- ADD-lite as topology deformation field.

Benchmark:

- against Markov/HMM regime allocators,
not merely momentum.

Core thesis:

PolygonEye's edge lies not in estimating transition probabilities better,
but in detecting:

**degeneration of future maneuverability under collapsing
ordering geometry.**

